What Really Predicts a Hit? Artist History, Era Context and Shortcut Features in Hit Song Prediction

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***Abstract*—Hit song prediction from Spotify audio descriptors has become a popular benchmark, but reported accuracies are rarely compared on the same data, rarely tested for significance, and rarely checked for shortcuts created by the way the datasets were built. We study two simple, non-audio signals: a *leakage-free artist history* feature (the number of Billboard Hot 100 songs the artist placed in strictly earlier periods) and the song's *era*. On the 40,560-song Spotify Hit Predictor dataset, adding both features to the 13 Spotify audio descriptors raises Random-Forest accuracy from 0.782 to 0.832 (+5.0 pts, 25/25 folds of repeated cross-validation, corrected $p = 7 \times 10^{-17}$). In a faithful re-implementation of the only academic study on this dataset, our feature improves its models by +2.1 to +3.2 pts on every one of 50 random splits. Because this benchmark excludes the artists of hits from its negative class, we validate the feature on the peer-reviewed ISMIR 2019 dataset, whose negatives are not selected by artist: it adds +2.4 pts and reaches 0.763, above the best published result (0.750), and it still adds +2.6 pts when hits and non-hits are matched year by year, a control under which release year and decade become useless. These gains hold when no artist appears in both training and test folds; a cold-start breakdown shows that artist history mainly helps established artists, while era context helps new ones. A systematic audit also shows that several gains are artefacts or noise: song titles (+6.7 pts) encode how the benchmark was assembled, and a 185-dimensional librosa representation that looked significant on a single split (+3.8 pts) is not significant under repeated cross-validation. Spectral descriptors of 30-second previews add +1.4 pts on the whole benchmark, but the gain vanishes on a genre-matched sample: the benchmark's negatives are chosen by genre, and the artist's genre alone lifts accuracy to 0.928. Treated as time series, chart data add nothing before release (market-trend features carry only the era), but once a song has charted its first three weeks predict a top-10 peak with an AUC of 0.89. All numbers are reproduced by released scripts.**

***Index Terms*—hit song prediction, music information retrieval, feature engineering, data leakage, shortcut learning, reproducibility**

I. Introduction

Whether a song will become a hit is a question with obvious commercial value and a long, contested research history. Early work concluded that "hit song science is not yet a science" [1], later work argued the opposite on other data [2], and the field has since moved from hand-crafted audio descriptors to convolutional networks on spectrograms [3] and multimodal models combining audio, lyrics and metadata [4]–[6].

A large share of recent studies relies on the audio descriptors exposed by the Spotify Web API and on labels derived from the Billboard Hot 100 chart [7]–[10]. Three weaknesses make these results hard to interpret. First, papers report accuracies on *different* datasets, so numbers such as 86 %, 88 % or 89 % are not comparable. Second, gains are usually measured on a single train/test split without a significance test. Third, the datasets are assembled with sampling rules that a model can exploit: a feature may separate hits from non-hits because of how the negative class was chosen rather than because of anything that makes a song successful, an instance of data leakage [11] and shortcut learning [12].

This paper studies two cheap, non-audio signals—the artist's prior chart history and the song's era—under an evaluation protocol designed to address these three weaknesses. Artist history is not a new idea. Interiano et al. [13] add a "superstar" indicator (whether the artist charted in the recent past) to acoustic features of 500,000 UK releases and find that it raises Random-Forest accuracy, and Middlebrook and Sheik [7] list an artist past-performance field among chart-derived fields, although the inputs of their final model are not stated. Neither study tests the gain for significance, compares it with prior models on the same data, or asks whether it reflects the way the negative class was sampled. What we add is a version restricted to strictly earlier periods, a significance-tested measure of its gain on the same data as prior work, and a check on an independent dataset of whether that gain is a genuine signal or a sampling artefact. Our contributions are:

1) A leakage-free formulation of *artist history* which, together with era, improves every classifier we tested on the Spotify Hit Predictor benchmark (Sec. V-A).
2) A *head-to-head* comparison with the only academic work on that benchmark, using its exact protocol, in which our feature improves its models on 50 out of 50 random splits (Sec. V-B).
3) An *external validation* on the peer-reviewed IS-

MIR 2019 dataset, including a year-matched control that separates a genuine signal from a sampling artefact, and in which our feature exceeds the best published accuracy (Sec. V-C).

4) A *robustness* analysis with artist-grouped cross-validation and a cold-start breakdown by whether the artist had an earlier hit (Sec. V-D).
5) An *audit* showing which apparent gains do not survive: title-based features that encode dataset construction, and a rich audio representation whose single-split gain disappears under repeated cross-validation (Sec. V-E), and a genre control showing that a significant spectral-feature gain on the whole benchmark is a genre shortcut (Sec. V-F).
6) A *time-series* analysis of the Hot 100: components and forecasting of chart dynamics, market-trend features before release, and early chart trajectories after release (Sec. V-G).

II. Related Work

Audio-based hit prediction. Pachet and Roy [1] found that acoustic and human-annotated descriptors could not predict popularity, while Ni et al. [2] and Herremans et al. [14] reported above-chance results on chart data. Yang et al. [3] revisited the problem with convolutional networks on audio. Zangerle et al. [4] combined low- and high-level Essentia descriptors [15] of Million Song Dataset tracks [16] in a wide-and-deep network and reached 75.04 % accuracy on a balanced set of 11,664 songs, noting that adding the release year improved every configuration.

Artist reputation. Interiano et al. [13] analyse more than 500,000 UK songs (1985–2015) and show that a superstar indicator improves the prediction of chart success over acoustic features alone. Salganik et al. [17] showed experimentally that success in cultural markets depends on social influence and cumulative advantage, which is one reason why an artist's past success predicts future success.

Spotify descriptors and Billboard labels. Middlebrook and Sheik [7] matched about 24,000 Spotify tracks (1985–2018) to the Hot 100 and reported 87.7 % test accuracy with a Random Forest; their feature list includes chart-derived fields and an artist past-performance feature. SpotHitPy [8] reported about 86 % on roughly 18,500 songs. Zhao et al. [9] added engineered metadata and lyrics features for Top-10 prediction (89.1 % with a Random Forest). These numbers come from different datasets and label definitions. The Spotify Hit Predictor dataset [18] is a public benchmark of decade-balanced hits and non-hits (1960–2019); the only academic study we found on it is an MSc thesis by Du [10], which reports 84 % (2000s) and 82 % (2010s) with one Random Forest per decade.

Multimodal and popularity prediction. Martín-Gutiérrez et al. [5] released SpotGenTrack and predicted Spotify popularity with an end-to-end multimodal network; DeepHits [6] combines log-Mel CNNs, lyric embeddings, artist popularity and release year on the same data; Rompolas et al. [19] use social-media sentiment.

TABLE I
Datasets. "Hit" definitions differ; accuracy is only comparable within a row.

Dataset	Songs	Hits	Label / negatives
Spotify Hit Predictor [18]	40,560	49.4 %	Hot 100 in decade / rule-based
audio subset (Deezer clips)	3,562	52.0 %	same
ISMIR 2019 [4] (balanced)	11,662	50.0 %	Hot 100 / random MSD
SpotGenTrack [5]	101,909	19.4 %	popularity ≥ 55

Evaluation pitfalls. Dietterich [20] recommends McNemar's test for a single split, and Nadeau and Bengio [21] show that naive t-tests over overlapping resamples are anti-conservative and propose a corrected variance. Leakage [11] and shortcut learning [12] describe models that exploit cues unrelated to the intended task. In music information retrieval, Sturm [22] showed that systems can reach high scores through confounds in the ground truth ("horses"), and Flexer [23] showed that letting songs by the same artist appear in training and test sets inflates genre-classification accuracy; our artist-grouped cross-validation applies this artist filter to hit prediction. To our knowledge, no hit-prediction study on Spotify descriptors reports both a significance-tested head-to-head on the same data and an audit of construction artefacts.

III. Data

Spotify Hit Predictor. Six decade files (1960s–2010s) list 13 Spotify audio descriptors (danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, tempo, duration, time signature), two analysis fields (`chorus_hit`, `sections`), the artist, the title and a binary target [18]. After removing duplicate track URIs, 40,560 songs remain. A song is a hit if it appeared on the weekly Hot 100 in that decade. Crucially, the dataset documentation states that a non-hit must satisfy all of: the track is not in that decade's hit list, *its artist is not in that decade's hit list*, it belongs to a "non-mainstream and/or avant-garde" genre, and that genre has no song in the hit list. Each decade file is therefore balanced (48.3–49.9 % hits), and the negative class is selected partly *by artist* and *by genre*. We return to this in Sec. VI.

Audio subset. For 3,562 of these songs a 30-second preview was first retrieved from Deezer (extended to 35,370 songs in Sec. V-E), from which we extracted 31 spectral/rhythmic descriptors ("FFT") and 185 richer librosa [24] descriptors (20 MFCCs and their first and second derivatives, chroma, tonnetz, spectral contrast, centroid, bandwidth, roll-off, flatness, zero-crossing rate, RMS and onset/tempogram statistics, each summarised by mean and standard deviation).

ISMIR 2019. The release of Zangerle et al. [4] provides Essentia low- and high-level descriptors for 5,832 Hot 100 hits and 89,235 non-hits from the Million Song Dataset.

We flatten all numeric descriptors (592 after removing file metadata such as bit rate and codec), keep the 5,831 hits with complete descriptors and, like the original study, draw an equal number of non-hits. Hits have a median release year of 1987 and non-hits of 2001, a fact that matters below.

SpotGenTrack. 101,909 Spotify tracks with 13 descriptors, 207 pre-extracted low-level audio features and a popularity score [5]; we label tracks with popularity ≥ 55 as hits (19.4 %). Because the classes are imbalanced, we report ROC-AUC alongside accuracy for this dataset.

A fourth candidate, a Billboard year-end chart file (2000–2023) augmented with Spotify features, was discarded: 2,911 of its 3,397 rows have no audio descriptors.

IV. Method

A. Features

Audio. The 13 Spotify descriptors (Hit Predictor), the 592 Essentia descriptors (ISMIR) or the 13 Spotify descriptors (SpotGenTrack) form the baseline of each dataset.

Era. On the Hit Predictor we one-hot encode the decade; on ISMIR and SpotGenTrack we use the decade (or the release year) as a number.

Artist history. Let $a(s)$ be the normalised primary artist of song s (accents, punctuation and "feat./&/with" collaborators removed) and $t(s)$ its period (decade on the Hit Predictor, release year elsewhere). With $\mathcal{H}$ the set of reference hits,

$$h(s) = \log\bigl(1 + \bigl|\{\, s' \in \mathcal{H} : a(s') = a(s),\ t(s') < t(s) \,\}\bigr|\bigr). \tag{1}$$

The strict inequality guarantees that no information from the song's own period or from the future is used. On the Hit Predictor, $\mathcal{H}$ is the full weekly Hot 100 history (distinct titles per artist and decade); on ISMIR and SpotGenTrack it is the set of positive songs of the dataset itself. On the Hit Predictor, 52.1 % of songs have an artist that can be matched to the chart history; this includes 99.99 % of the hits, and almost all unmatched songs are flops by artists who never charted (Sec. V-F).

Rich audio. Standardised and reduced by PCA to 40 components; scaling and PCA are fitted on the training part of each fold only.

B. Models

The main model is the Random Forest [25] configuration of Middlebrook and Sheik [7] (80 trees, 8 candidate features per split, Gini), kept frozen throughout: we do not tune hyperparameters, so that gains come from the features and remain comparable with the published configuration. Robustness checks use XGBoost [26], logistic regression, k-NN and an RBF support vector machine. External datasets use a Random Forest with 200 trees and $\sqrt{d}$ candidate features, as in our original external scripts.

TABLE II
Spotify Hit Predictor (40,560 songs), frozen Random Forest. Canonical split (2,000 test songs, McNemar) and 5×5 cross-validation (mean, 95 % CI, corrected t-test, folds improved out of 25).

	Canonical		Cross-validation		
Features	Acc.	p	Acc. [95 % CI]	AUC	p (wins)
Audio (13)	.773	–	.782 [.780,.784]	.859	–
+ decade	.796	$3{\cdot}10^{-4}$	.802 [.800,.804]	.880	10^{-10} (25)
+ artist	.800	$6{\cdot}10^{-4}$	.810 [.808,.813]	.900	10^{-11} (25)
+ decade + artist	**.822**	$2{\cdot}10^{-9}$	**.832 [.830,.834]**	**.918**	$7{\cdot}10^{-17}$ (25)
Decade + artist only	.668	–	.670 [.669,.672]	.685	–

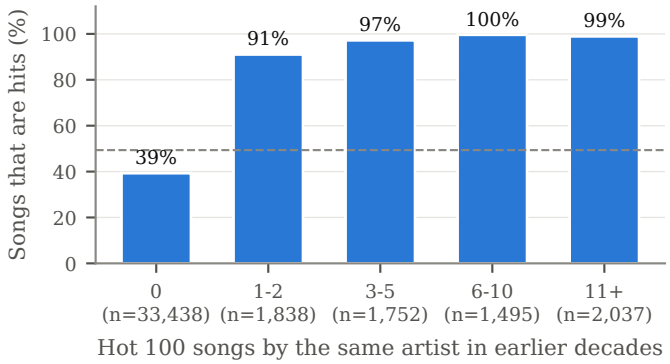


Fig. 1. Share of hits by the number of Hot 100 songs the artist placed in earlier decades (Spotify Hit Predictor, 40,560 songs). Dashed line: overall hit share (49 %).

C. Evaluation protocol

We report two estimates for every comparison. The *canonical* estimate reproduces the split of our earlier experiments (seed 42; 20,000 training and 2,000 test songs on the full benchmark, 75/25 elsewhere) and is compared with McNemar's test [20]. The *cross-validated* estimate uses repeated stratified 5-fold cross-validation (5 repeats unless stated), reports the mean and a 95 % confidence interval, and compares configurations fold by fold with the corrected resampled t-test of Nadeau and Bengio [21]. All learned transformations are inside the cross-validation loop. Title features are produced out-of-fold: a TF-IDF + logistic-regression model is trained within the training part and its predicted probability is used as a single feature.

V. Results

A. Main result on the Spotify Hit Predictor

Table II shows that each feature helps on its own and that their gains are nearly additive. The canonical numbers reproduce our earlier single-split experiments exactly; cross-validation places all accuracies about one point higher and gives very narrow intervals. The combined gain is +5.04 pts and is positive in all 25 folds. It is not bought by trading recall for precision: on the hit class, precision rises from 0.758 to 0.828 and recall from 0.819 to 0.833, so F1 rises from 0.787 to 0.831 (+4.3 pts, 25/25, $p = 7 \times 10^{-15}$). Without audio the two features reach

TABLE III
CROSS-VALIDATED ACCURACY PER DECADE (RANDOM FOREST).

Features	60s	70s	80s	90s	00s	10s
Audio (13)	.743	.750	.770	.806	.836	.816
+ decade	.764	.763	.794	.840	.851	.833
+ artist	.763	.784	.812	.831	.860	.842
+ decade + artist	**.788**	**.796**	**.836**	**.864**	**.873**	**.868**

TABLE IV
ROBUSTNESS TO THE CLASSIFIER (5-FOLD CV, CORRECTED t-TEST; ALL 5 FOLDS IMPROVE).

Classifier	Audio (13)	+ decade + artist	Δ	p
Random Forest [25]	.780	.832	+5.2	$2{\cdot}10^{-5}$
XGBoost [26]	.788	**.845**	+5.7	$3{\cdot}10^{-7}$
Logistic regression	.727	.782	+5.5	$4{\cdot}10^{-5}$
k-NN ($k = 25$)	.750	.796	+4.6	$8{\cdot}10^{-6}$
SVM (RBF)	.773	.828	+5.5	$3{\cdot}10^{-5}$

only 0.670, so they complement rather than replace the descriptors. The relation between artist history and the label is strongly nonlinear (Fig. 1): 39.3 % of the 33,438 songs whose artist had no earlier hit are hits, against 91.0 % for one or two earlier hits and 97.2–99.6 % beyond.

The improvement holds in every decade (Table III), from +3.7 pts in the 2000s to +6.6 pts in the 1980s. Since each decade file is balanced, the decade indicator carries no information on its own; its gain can only come from interactions with the audio descriptors, i.e. from era-specific decision rules.

Robustness to the classifier. The gain is of the same size for five classifiers of very different nature (Table IV), so it comes from the features rather than from the model. XGBoost is the strongest on average, but on the canonical split it ties with the frozen Random Forest (0.820 vs 0.822, McNemar $p = 0.87$). TreeSHAP attributions of the XGBoost model rank artist history first (mean $|\phi| = 1.08$), the six decade indicators second (0.86 summed) and instrumentalness third (0.73), ahead of every other descriptor.

Future decades. To test whether the gain survives a temporal shift, we train on all earlier decades and test on a later one, with no song from the test decade seen in training. Artist history improves the Random Forest by +2.5 pts on the 1990s (0.736 → 0.761), +1.9 pts on the 2000s (0.805 → 0.824) and +3.4 pts on the 2010s (0.780 → 0.814; McNemar p between 10^{-7} and 10^{-21}), with similar gains for XGBoost. The decade indicator cannot be used here, since the test decade never appears in training.

B. Head-to-head with prior work on the same benchmark

Du [10] trained one Random Forest (450 trees, $m_{\text{try}} = 2$, at most 30 leaves) and one logistic regression per decade on the 2000s and 2010s, with a single 70/30 split. Because the original R random seed cannot be replayed, we repeat the split with 50 seeds. Our re-implementation lands on or slightly above the published figures (Table V, Fig. 2), which validates the protocol. Adding artist history to *their* models, with *their* inputs, improves accuracy by +2.1 to +3.2 pts on every split, and the improved models exceed the published accuracy on 100 % of the splits. The decade feature is constant within a per-decade model and is therefore not used here.

TABLE V
HEAD-TO-HEAD WITH DU [10] USING THEIR PROTOCOL (ONE MODEL PER DECADE, 13 AUDIO + `CHORUS_HIT` + `SECTIONS`, 70/30 SPLIT) OVER 50 RANDOM SPLITS: MEAN ACCURACY (STD). "WINS" COUNTS SPLITS WHERE ADDING ARTIST HISTORY HELPS.

Dec.	Model	Published	Re-impl.	+ artist	Wins / p
00s	RF (their cfg.)	.84	.846 (.009)	**.867** (.009)	50 / $9{\cdot}10^{-8}$
	LR (their cfg.)	.82	.821 (.007)	**.848** (.007)	50 / $2{\cdot}10^{-9}$
	RF (frozen)	–	.852 (.008)	**.875** (.006)	50 / 10^{-7}
10s	RF (their cfg.)	.82	.829 (.008)	**.861** (.008)	50 / $4{\cdot}10^{-11}$
	LR (their cfg.)	.79	.807 (.009)	**.835** (.008)	50 / $6{\cdot}10^{-6}$
	RF (frozen)	–	.841 (.008)	**.873** (.008)	50 / $9{\cdot}10^{-10}$

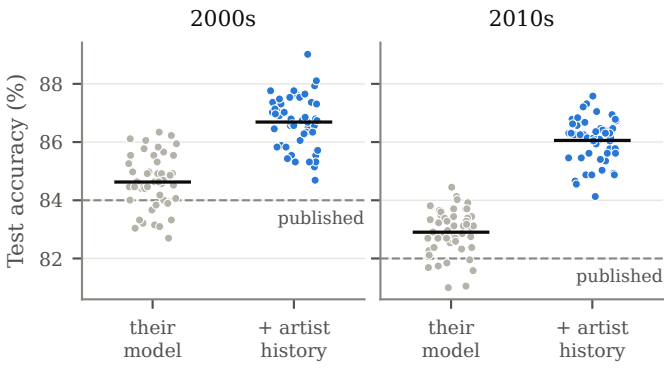


Fig. 2. Head-to-head with Du [10]: accuracy of their Random Forest over 50 random 70/30 splits (dots; bar = mean), without (gray) and with (blue) artist history. Dashed line: their published accuracy.

C. External validation

ISMIR 2019. (Table VI, Fig. 3.) With random balanced negatives (protocol A, as in the original study), release year alone reaches 70.2 %—as much as 592 audio descriptors—because hits and non-hits come from different eras. Audio + year, which mirrors the inputs of the best model of Zangerle et al., reaches 73.9 %; adding artist history raises it to 76.3 % (+2.4 pts, 10/10 folds, $p = 7{\times}10^{-8}$), above the published 75.04 %. Protocol B removes the era shortcut by drawing, for every hit, a non-hit from the same release year. Year and decade then become useless ($p > 0.5$, and year alone falls below chance), while artist history still adds +2.6 pts ($p = 3 \times 10^{-5}$, 10/10 folds). This is the cleanest estimate of the feature's value: the negatives are not selected by artist and cannot be separated by date.

SpotGenTrack. Here the label is Spotify popularity rather than a chart, and 80.6 % of tracks are negatives, so accuracy is dominated by the majority class: the 13

TABLE VI

ISMIR 2019 DATA, RANDOM FOREST ON 592 ESSENTIA DESCRIPTORS, 5×2 CROSS-VALIDATION. A: BALANCED RANDOM NEGATIVES (AS IN [4]). B: NEGATIVES MATCHED TO HITS YEAR BY YEAR. p: CORRECTED t-TEST VS AUDIO; ALL 10 FOLDS IMPROVE WHERE SIGNIFICANT.

	A: random negatives		B: year-matched	
Features	Acc. [95 % CI]	p	Acc. [95 % CI]	p
Audio (Essentia)	.708 [.700,.717]	–	.682 [.677,.688]	–
+ year	.739 [.734,.744]	$2{\cdot}10^{-4}$	.681 [.674,.688]	.66
+ artist	.733 [.727,.740]	$2{\cdot}10^{-5}$	**.708** [.702,.714]	$3{\cdot}10^{-5}$
+ year + artist	**.763** [.758,.768]	$7{\cdot}10^{-8}$	**.708** [.702,.715]	$3{\cdot}10^{-5}$
Year alone	.702 [.697,.707]	–	.448 [.444,.451]	–

Best published [4]: 75.04 % (wide-and-deep, audio + year, 5-fold CV)

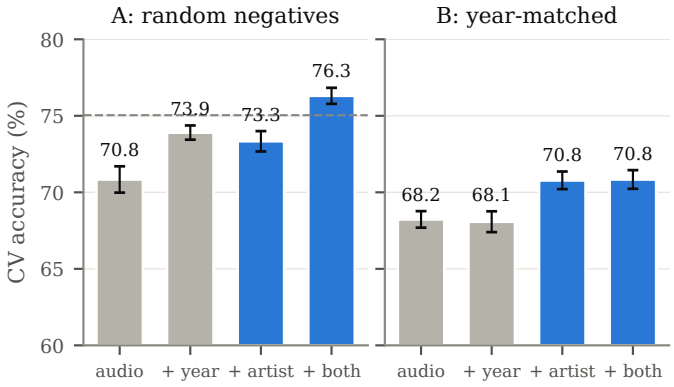


Fig. 3. ISMIR 2019 data, cross-validated accuracy with 95 % CI. Gray: without artist history; blue: with it ("both" = year + artist). Dashed line in A: best published accuracy [4]. Matching years (B) removes the gain of release year but not that of artist history.

descriptors score 0.812, barely above always predicting "non-hit". Ranking quality tells the real story (Table VII): artist history raises ROC-AUC from 0.690 to 0.762 (+7.2 points), and with decade to 0.777. On the hit class, the 13 descriptors find only 8.3 % of the hits (F1 0.147); with decade and artist history, recall rises to 25.2 % at unchanged precision (0.62) and F1 to 0.358 (5/5 folds, $p = 7 \times 10^{-7}$), while the 207 low-level audio features add nothing on top of artist history. A single 75/25 split gives the same picture (accuracy 0.810 → 0.822 and ROC-AUC 0.688 → 0.760 with artist history, McNemar $p = 6 \times 10^{-13}$).

D. Unseen artists and cold start

Artist-grouped cross-validation. With random folds, songs by the same artist can fall in both the training and the test part, so a model could recognise an artist from the audio. We therefore repeat every protocol with stratified group k-fold, grouping songs by normalised artist so that no test artist is seen in training (Table VIII, top). Accuracies drop by up to one point, a small artist-identity effect in the audio descriptors, but the gains remain: +5.0 pts for decade and artist on the Hit Predictor (unchanged, 25/25 folds, $p = 10^{-13}$) and +2.9 pts for

TABLE VII

SPOTGENTRACK (101,909 TRACKS, 19.4 % HITS; ALWAYS PREDICTING "NON-HIT" SCORES 0.806). 5-FOLD CV, CORRECTED t-TEST VS THE 13 DESCRIPTORS.

Features	Acc. [95 % CI]	ROC-AUC	p (acc.)
Spotify descriptors (13)	.812 [.811,.813]	.690	–
+ decade	.812 [.811,.813]	.712	.66
+ artist	.823 [.821,.826]	.762	10^{-3}
+ decade + artist	**.825** [.823,.827]	**.777**	$3{\cdot}10^{-4}$
+ artist + low-level audio	.825 [.823,.828]	.758	$2{\cdot}10^{-4}$

TABLE VIII

ROBUSTNESS. TOP: CROSS-VALIDATED ACCURACY WITH RANDOM STRATIFIED FOLDS VS FOLDS GROUPED BY ARTIST (NO TEST ARTIST SEEN IN TRAINING). ERA = DECADE (HIT PREDICTOR) OR RELEASE YEAR (ISMIR). BOTTOM: GAIN OVER AUDIO IN GROUPED CV, ON SONGS WHOSE ARTIST HAD NO EARLIER HIT (NEW) OR AT LEAST ONE (KNOWN). NEW/KNOWN SONGS: 33,438/7,122 (HIT PREDICTOR), 7,480/4,182 (ISMIR A), 7,269/4,393 (ISMIR B).

Data	Folds	Audio	+ era	+ artist	+ both
Hit Predictor	random	.782	.802	.810	.832
	by artist	.772	.792	.802	**.822**
ISMIR A	random	.708	.739	.733	.763
	by artist	.709	.739	.730	**.759**
ISMIR B	random	.682	.681	.708	.708
	by artist	.677	.678	**.705**	**.705**

	Hits (%)	New artists: Δ (p)		Known: Δ (p)
	new/known	+ era	+ artist	+ artist
Hit Predictor	39 / 97	+2.2 ($2{\cdot}10^{-9}$)	+0.4 (.17)	+15.0 ($3{\cdot}10^{-14}$)
ISMIR A	36 / 75	+3.4 ($6{\cdot}10^{-5}$)	+0.8 (.08)	+4.4 (.003)
ISMIR B	37 / 72	+0.1 (.79)	+1.4 (.002)	+5.3 ($8{\cdot}10^{-5}$)

artist history in the year-matched ISMIR protocol (10/10, $p = 9 \times 10^{-7}$). In protocol A the gains lose at most half a point, and audio, year and artist still reach 0.759 [0.752, 0.766], above the published 75.04 %.

Cold start. Table VIII (bottom) splits the grouped test folds by whether the artist had an earlier hit. On the Hit Predictor, artist history does not help new artists (+0.4 pts, $p = 0.17$): its whole gain comes from the 17.6 % of songs by established artists, 96.6 % of which are hits, which quantifies the benchmark effect discussed in Sec. VI. The decade, in contrast, helps new artists (+2.2 pts, $p = 2 \times 10^{-9}$). On ISMIR, where songs by established artists are hits only 72–75 % of the time, artist history also tends to help new artists in the year-matched protocol (+1.4 pts, $p = 0.002$, 10/10 folds), presumably because an empty chart history is itself informative when negatives are not selected by artist.

E. Audit: which gains are not real

Titles. A title model trained out-of-fold adds +6.7 pts, more than any musical feature. Titles do not make songs popular; they reveal the construction of the benchmark, whose non-hits are drawn from niche genres and catalogue releases whose titles differ systematically from Hot 100 singles. We reject this feature despite its significance.

TABLE IX
AUDIT OF CANDIDATE GAINS. SINGLE SPLIT (MCNEMAR) VS 5×5 CROSS-VALIDATION (CORRECTED t-TEST). TOP: FULL BENCHMARK. BOTTOM: 3,562-SONG AUDIO SUBSET.

Added to audio	Split Δ (p)	CV Δ	CV p	Verdict
Song title (TF-IDF)	+6.9 ($4{\cdot}10^{-14}$)	+6.7	$4{\cdot}10^{-20}$	artefact
`chorus_hit+sections`	+0.6 (.32)	−0.0	.82	no effect
Title length	+0.05 (1.0)	+0.2	.28	no effect
Rich librosa (PCA 40)	+3.8 (.002)	+0.9	.15	not significant
FFT descriptors (31)	+4.4 (.001)	+2.3	.007	small
Decade	+1.1 (.27)	+1.0	.03	small
Artist	+3.4 (.010)	+3.8	$2{\cdot}10^{-6}$	robust
Artist + rich	+6.7 (10^{-6})	+4.9	10^{-8}	= artist
Decade + artist	+3.6 (.006)	+5.1	10^{-8}	robust

Single split vs cross-validation. On the small audio subset, the 185 rich librosa descriptors appeared to add +3.8 pts with $p = 0.002$ on the canonical split, but the cross-validated gain is +0.9 pts and is not significant. The same holds for the choice of the PCA dimension (5 to 100 components are within noise) and for the knock-out of any descriptor family (all within ±0.1 pts). In a block ablation of the full subset model, only removing artist history causes a significant drop (−3.3 pts, $p = 10^{-5}$). The `chorus_hit` and `sections` fields, often suspected of leakage, add nothing.

FFT and the best audio-subset model. The 31 FFT descriptors are the only extra audio representation whose gain survives cross-validation, and it persists on top of artist history: audio + artist + FFT reaches 0.821 against 0.801 for audio + artist (+2.0 pts, 23/25 folds, $p = 0.008$). Adding decade to that model gives the best subset accuracy, 0.824, but the increment is not significant (+0.3 pts, $p = 0.50$), and decade adds nothing to FFT alone (0.786 vs 0.785): the spectral descriptors already carry the era. Conversely, artist history still adds +3.5 pts on top of FFT ($p = 4 \times 10^{-5}$). Fig. 4 shows every configuration we evaluated on this subset.

FFT on the whole benchmark. We then retrieved Deezer previews for every song we could match: 35,370 of the 40,560 songs (87.2 %). Deezer finds hits more often than flops (94.0 % vs 80.6 %), so the covered set is 53.2 % hits and every comparison below uses the same 35,370 songs. With 5×5 cross-validation, adding FFT to the main model (audio + decade + artist) raises accuracy from 0.825 to 0.839 (+1.4 pts, 25/25 folds, $p = 4 \times 10^{-9}$; AUC 0.911 → 0.921). The whole gain comes from songs by new artists, 0.789 → 0.807 (+1.8 pts, 25/25, $p = 6 \times 10^{-9}$), while songs by known artists stay at 0.977 (−0.1 pts, n.s.). Unlike on the small subset, decade remains useful once FFT is present: audio + artist + FFT reaches only 0.829 (+0.4 pts over the main model, $p = 0.04$). FFT thus appears to address the cold-start weakness of artist history, but the benchmark's flops come from non-mainstream genres by construction, and spectral descriptors can recognise genre. Sec. V-F shows that this gain does not survive a genre control.

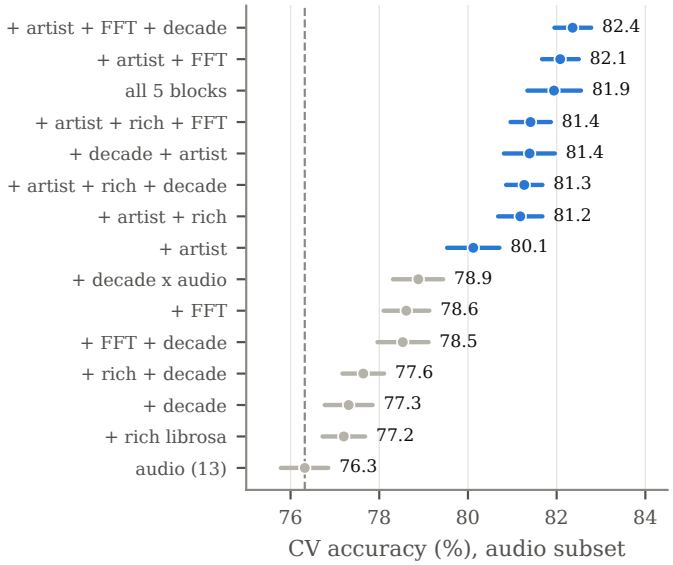


Fig. 4. Every configuration evaluated on the 3,562-song audio subset, 5×5 cross-validated accuracy with 95 % CI, sorted. Blue: with artist history; gray: without. Dashed line: the 13 Spotify descriptors alone.

F. Further experiments

Genre control for FFT. We attach Spotify artist genres from the SpotGenTrack release [5] by normalised artist name (20,610 songs with both a preview and a genre) and map the tags to 16 families. The benchmark's construction is plainly visible: Latin, classical, jazz, "world" and metal artists account for 50 % of the flops and 1.9 % of the hits, while rock, pop, R&B, hip-hop and country account for 93 % of the hits. Genre is also partly audible: a Random Forest recovers the family from the 13 Spotify descriptors with 47.4 % accuracy, from the 31 FFT descriptors with 49.9 % and from both with 53.3 % (majority class 26.1 %). Three results follow (Table X). (a) Genre families as features raise the main model from 0.828 to 0.928 (+10.1 pts, 25/25, $p = 8 \times 10^{-23}$): on this benchmark, knowing the artist's genre is worth more than any audio representation. (b) Once genre is known, FFT *lowers* accuracy (−1.1 pts, 0/25, $p = 5 \times 10^{-7}$), also for new artists (−1.4 pts). (c) On a genre-matched sample (as many hits as flops inside every family; 7,554 songs), FFT adds nothing to the main model (+0.1 pts, 13/25, $p = 0.84$; new artists +0.1 pts, $p = 0.80$). FFT still improves the 13 Spotify descriptors alone on that sample (+2.5 pts, 25/25, $p = 5 \times 10^{-4}$), but the main model, without FFT, still beats audio + FFT by 4.3 pts ($p = 2 \times 10^{-8}$). We conclude that the full-benchmark FFT gain is a genre shortcut, whereas the gain of artist history survives the genre control. Spotify assigns artist genres retrospectively, so we use them only as a control, never as a predictive feature.

Artist resolution. The 52.1 % match rate is not a matching problem: 99.99 % of the hits are found in the

TABLE X
GENRE CONTROL FOR FFT (5×5 CV). LEFT: SONGS WITH A PREVIEW AND A SPOTIFY GENRE. RIGHT: GENRE-MATCHED SAMPLE. ACCURACY OVERALL AND ON SONGS BY NEW ARTISTS.

	All (20,610)		Matched (7,554)	
Model	Acc.	new art.	Acc.	new art.
main (audio+decade+artist)	.828	.780	.790	.749
+ FFT	.841	.798	.791	.750
+ genre	.928	.911	–	–
+ genre + FFT	.917	.897	–	–
FFT vs main, Δ (p)	+1.3 (10^{-4})		+0.1 (.84)	
FFT vs main+genre, Δ (p)	−1.1 (5×10^{-7})		–	

TABLE XI
ROLLING ORIGIN: TRAIN ON ONE DECADE, TEST k DECADES LATER (RANDOM FOREST, 35,370 SONGS WITH A PREVIEW; $k = 0$: 5-FOLD CV INSIDE EACH DECADE). MEAN ACCURACY OVER THE AVAILABLE DECADE PAIRS.

k	pairs	audio	+ artist	+ FFT	+ artist + FFT
0	6	.809	.837	.831	**.859**
1	5	.765	.796	.782	**.810**
2	4	.728	**.775**	.736	**.775**
3	3	.689	**.740**	.673	.728
4	2	.666	**.734**	.661	.718
5	1	.664	**.754**	.599	.706

chart history up to their own decade, and the unmatched songs are flops whose artists never charted (2.8 % of flops are matched). Two alternatives give no gain: matching the full credited name before the primary artist ("Simon & Garfunkel" rather than "Simon") lowers accuracy (0.827 vs 0.832, −0.5 pts, $p = 4 \times 10^{-4}$), and crediting featured appearances changes nothing (+0.1 pts, $p = 0.39$). Canonical identifiers would avoid homonyms but cannot raise the match rate.

How fast does a model age? We train on a single decade and test on every later one (Table XI, songs with a preview). Audio-only accuracy falls from 0.809 within a decade to 0.765 one decade later and 0.689 three decades later. Artist history slows the decay, and its advantage grows with the gap (+2.8 pts within a decade, +5.1 pts three decades later). FFT behaves in the opposite way: it helps within a decade (+2.2 pts) but hurts from three decades on, consistent with descriptors that encode era-specific production and genre rather than lasting appeal. The cells average 6 to 1 train/test pairs and are not tested for significance.

Lyrics. SpotGenTrack provides six lexical descriptors of the lyrics (syllables per word, words per sentence, sentence count, word count, sentence similarity, vocabulary richness) for 79,420 tracks with lyrics (21.5 % hits; 5×1 CV as in Sec. V-C). They improve the 13 Spotify descriptors (AUC 0.678 → 0.729, $p = 3 \times 10^{-4}$) and still add on top of decade and artist history (accuracy 0.808 → 0.814, 5/5 folds, $p = 0.002$; AUC 0.774 → 0.797, $p = 2 \times 10^{-5}$). The gain is modest in accuracy because of the class imbalance, but it is the only content feature in this study that adds to artist history on a dataset whose negatives are not selected by genre.

G. Time series: chart dynamics

The chart as a time series. The weekly Hot 100 (3,545 charts, 1958–2026) also forms time series. The monthly mean number of debuts per chart week (1959–2025) is mostly trend: an STL decomposition [27] gives a trend strength of 0.72 and a seasonal strength of 0.11 [28]. Debuts fall from 11.0 per week in 1959 to 5.5 in 2001 and return to 10.9 in 2025, after the chart started counting streams; the largest residual is a rule change (60 debuts in the chart of 5 December 1998, when songs not released as singles became eligible). The weekly number of holiday songs on the chart is almost purely seasonal: its periodogram peaks at 52.1, 26.1, 17.4 and 13.0 weeks, and a Fourier regression fitted on the streaming era ($K = 20$ harmonics) explains 94 % of its variance. Forecasting monthly debuts 12 months ahead under rolling-origin evaluation (11 origins, 2015–2025), ARIMA$(1, 1, 2)$ [29] and ARIMA with Fourier terms (MASE 1.82) and damped Holt-Winters (1.85) are best and indistinguishable ($p = 0.97$); they beat the seasonal naive forecast (2.32) at 10 of 11 origins ($p = 0.006$). A seasonal ARIMA (1.89) and gradient boosting or random forests on lag and Fourier features (1.94, 1.99) do worse: the series has little seasonality, and 800 points leave tree models little to learn.

Before release: market trends. We describe each Hit Predictor song by the distance of its standardised audio descriptors to the centroid of the previous decade's hits, and by its projection on the direction in which that centroid moved. On top of the main model these two features add +0.07 pts ($p = 0.66$); the 13 per-feature deviations add +0.46 pts (23/25, $p = 0.006$). On audio alone the trend features add +1.8 pts ($p = 7 \times 10^{-10}$), less than the decade alone (+2.0 pts): they carry era information, which the main model already has.

After release: early trajectories. A different question can be asked once a song has entered the chart: will it become a big hit? We take the 31,925 songs with a complete history (debut after the first chart and at least 52 weeks of follow-up; 16.4 % reach the top 10) and describe each by its positions in its first k weeks (101 if off the chart), best rank, slope and acceleration, with the year, month and the artist's earlier songs and earlier top-10 hits reached before the debut as context (Random Forest, 200 trees, 5×3 CV). Table XII shows that the trajectory dominates: three weeks give an AUC of 0.892 against 0.593 for the context alone, and context adds +0.03 to +0.05 on top (15/15 folds, $p < 10^{-6}$ for every k). The prediction is not limited to songs already at the top: among the songs not yet in the top 10 after three weeks, the AUC is 0.870, and a model trained on songs before 2000 reaches 0.852 on songs from 2000 onwards. As for online content [30],

TABLE XII
PREDICTING A TOP-10 PEAK FROM A SONG'S FIRST k WEEKS ON THE HOT 100 (31,925 SONGS, 16.4 % TOP-10; RANDOM FOREST, 5×3 CV). CONTEXT: YEAR, MONTH AND THE ARTIST'S EARLIER CHART HISTORY. LAST COLUMN: SONGS NOT YET IN THE TOP 10 AFTER k WEEKS.

		All songs			Not yet top 10
k	weeks	+ context	F1	context only	+ context
1	.700	.750	.401	.593	.720
2	.796	.850	.531	.593	.828
3	.845	**.892**	.610	.593	.870
4	.891	**.917**	.674	.593	.894

early popularity predicts later popularity. This task differs from that of Sec. V-A: every song has already charted, so it separates big hits from small ones and cannot be used before release. A second target, at least 20 weeks on the chart, depends on chart rules that remove older songs and is not reported.

VI. DISCUSSION

How much of the gain is real? The benchmark's negative class excludes the artists of that decade's hits. Our feature only counts *earlier* decades, so it is leakage-free in time, but an artist who charted before is likely to chart again and is therefore unlikely to appear as a negative by construction: flops have on average 0.037 earlier hits, against 3.21 for hits. The +5.0 pts on this benchmark, and the head-to-head gains of Table V, should be read as benchmark-specific. The ISMIR year-matched protocol, whose negatives are chosen neither by artist nor by date, gives a more conservative +2.6 pts; this is the number we consider transferable.

Era as a shortcut. On ISMIR, release year looks like a strong predictor (70.2 % alone, and +12–13 % in the original study) because hits are older than the non-hit pool. Once years are matched, it carries no information. On the Hit Predictor, the decade is uninformative alone and helps only through interactions, which is consistent with era-specific tastes but may also reflect decade-dependent genre rules of the negative class. Reported gains from date features should be accompanied by a year-matched control.

Genre as a shortcut. On the Hit Predictor, the artist's genre alone takes the main model to 0.928, and the apparent FFT gain disappears once genre is controlled (Sec. V-F). Any audio representation evaluated on this benchmark can score by recognising the excluded genres rather than by modelling what makes a song a hit, so audio gains on it should be reported together with a genre-matched control. Artist history passes that control; FFT does not. Within the audio content itself, therefore, no representation we tested adds to artist history and era once the construction shortcuts are controlled: on these data, what predicts a hit is mostly who sings it and when.

Single splits mislead. Several conclusions of our own earlier single-split analysis did not survive cross-validation. With a few thousand songs, an 891-song test set has a standard error of about 1.4 pts, so single-split gains of 3–4 pts on subsets are fragile even when McNemar's test is significant on that split.

Multiple comparisons. The paper reports about 60 p-values without correction for multiplicity. Every result our conclusions rest on remains significant after a Bonferroni correction over all of them ($p < 0.05/60 \approx 8 \times 10^{-4}$): the main gains of Tables II and IV, the head-to-head of Table V, the temporal hold-out, the ISMIR gains of artist history in both protocols, and the artist-grouped gains. The full-benchmark FFT gain on top of the main model also survives ($p = 4 \times 10^{-9}$) but is explained by genre (Sec. V-F); the genre-control results themselves ($p \leq 5 \times 10^{-7}$) survive as well, and so do the early-trajectory gains of Table XII ($p < 10^{-6}$). Results with $8 \times 10^{-4} < p < 0.05$ should be read as indicative only: the decade and FFT gains on the 3,562-song audio subset (including FFT on top of artist history), audio + artist + FFT against the main model on the whole benchmark ($p = 0.04$), the per-feature market-trend deviations ($p = 0.006$), the forecasting comparison of Sec. V-G ($p = 0.006$, 11 origins), the lyrics gains on SpotGenTrack (5×1 CV, $p = 0.002$ for accuracy), most single-split gains on that subset, the SpotGenTrack accuracy gain of artist history alone ($p = 10^{-3}$; with decade, $p = 3 \times 10^{-4}$), and the ISMIR cold-start gains of artist history for known artists in protocol A and new artists in protocol B.

Limitations. Artist history is matched by normalised name, not by a canonical identifier, so homonymous artists can be merged. Artist history does not help new artists on the Hit Predictor and helps them only modestly on ISMIR (+1.4 pts), and it partly measures cumulative advantage [17] rather than the intrinsic appeal of a song. We use classical classifiers on fixed descriptors rather than deep networks on raw audio [3], [6]: our question is which inputs carry signal under a controlled protocol, and the comparison with prior work is made at equal model; a learned representation could still capture what the descriptors miss. The comparison with Zangerle et al. uses a classifier rather than their regression on chart position and a different random draw of negatives; the comparison with Du uses re-drawn splits. Our audio previews are 30-second clips from a different service than the one that computed the Spotify descriptors.

VII. FUTURE WORK

Sec. V-F already runs the cheap versions of several follow-ups: artist matching (no gain available), a genre control for audio features, rolling-origin decay and lexical lyrics descriptors. What remains needs new data. (i) *Cold start*: artist history barely helps first-time artists on the benchmark, and the FFT gain for them was a genre shortcut; features such as label, collaborators, prior streams or

social media remain to be tested, with a genre-matched control. (ii) *Lyrics and emotion*: semantic and affective lyrics representations, and learned audio embeddings, evaluated with the same audit protocol. (iii) *Benchmark design*: a hit-prediction dataset whose negatives are sampled independently of artist, genre and date. Our genre-matched sample is a small, post-hoc approximation of it; a purpose-built one would let content features be compared without construction shortcuts. (iv) *Early trajectories*: Sec. V-G uses weekly chart positions only; daily streams and radio plays would allow the same prediction days after release.

VIII. Conclusion

A song's artist history is a simple, leakage-free signal that improves hit prediction across classifiers, beats the only prior model on the Spotify Hit Predictor benchmark in a head-to-head comparison, and transfers to an independent peer-reviewed dataset where it exceeds the best published accuracy and survives a year-matched control. These gains hold for artists never seen in training, although on the benchmark they come from established artists, whereas era context helps newcomers; spectral descriptors of a 30-second preview appear to help them too, but that gain disappears under a genre control. The same audit shows that larger gains—song titles, release dates, genre-driven audio features, rich audio on a single split—can come from how the data were built or from chance. Chart time series add nothing before release, but after release a song's first weeks are the strongest predictor we found. Knowing which gains are real matters more than the headline number.

Reproducibility

Every number in this paper is produced by the scripts in `repro/` (shared data loading and evaluation in `common.py`; one script per section) and written to `paper_results/`. Random seeds are fixed (42 unless stated).

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